BANGALORE SAHODAYA SCHOOLS COMPLEX ASSOCIATION

PRE-BOARD EXAMINATION -1 (2025 − 2026)

GRADE XII

Date: 10.12.2025 **Max Marks: 70**

Subject: COMPUTER SCIENCE (083) **Time: 3 hours**

GENERAL INSTRUCTIONS

- This question paper contains 37 questions.
- All questions are compulsory. However, internal choices have been provided in some questions. Attempt only one of the choices in such questions.
- The paper is divided into 5 Sections- A, B, C, D and E.
- Section A consists of 21 questions (1 to 21). Each question carries 1 Mark.
- Section B consists of 7 questions (22 to 28). Each question carries 2 Marks.
- Section C consists of 3 questions (29 to 31). Each question carries 3 Marks.
- Section D consists of 4 questions (32 to 35). Each question carries 4 Marks.
- Section E consists of 2 questions (36 to 37). Each question carries 5 Marks.
- All programming questions are to be answered using Python Language only.
- In case of MCQ, text of the correct answer should also be written.

Q.NO	SECTION -A (21 x 1=21 Marks)	MARKS
1	State if the following statement is True or False: math.pow(2,3) and 2**3 return same type of result.	1
2	State the correct output of the following code text1 = " Welcome " text2 = " To PythonWorld" result = text1.strip() + text2.replace("P", "p").lstrip() print(result.partition("To")) a) ('WelcomeTo ', 'PythonWorld', '') b) ['WelcomeTo ', 'To', ' pythonWorld'] c) ('Welcome', 'To', ' pythonWorld') d) ('Welcome', 'To', 'pythonWorld')	1
3	Consider the given expression: print(2**3**2 - 100//10%3 + 16//2**2) Which of the following will be the correct output? a) 511 b) 515 c) 520 d) 4096	1
4	If dict1 is a dictionary defined below, then which of the following will raise an exception? dict1={“Sam”:77, “Ram”:89, “Vijay”:100} a) dict1.get(“Vijay”) b) print(dict1([“Sam”, “Ram”]) c) dict1[“Ram”]=158 d) print(str(dict1))	1

5.	What will be the output of the following Python code?
	try:
	   a = 5
	   b = 2
	   print(a / (b - b))
	except Exception:
	   print("General Error")
	except ZeroDivisionError:
	   print("Arithmetic Error")
	else:
	   print("No Error")
	finally:
	   print("Done")
	
	a) Arithmetic Error Done b) General Error No Error c) General Error Done d) No Error Done
		1

6	Consider the given SQL Query: **SELECT department, AVG(salary)** **FROM employees** **WHERE salary > 50000** **GROUP BY department** **HAVING COUNT(*) >= 3** **ORDER BY AVG(salary);** Identify the correct sequence of execution for this query: a) WHERE → SELECT → GROUP BY → HAVING → ORDER BY b) GROUP BY → WHERE → HAVING → SELECT → ORDER BY c) SELECT → WHERE → GROUP BY → HAVING → ORDER BY d) WHERE → GROUP BY → HAVING → SELECT → ORDER BY	**1**
7	What will be the possible output of the following Python code? import random nums = list(range(10, 31, 5)) # [10, 15, 20, 25, 30] x = random.randrange(0, len(nums), 2) y = random.randrange(1, len(nums), 2) print(nums[y] - nums[x]) Which of the following cannot be the output? a) 5 b) 30 c) 15 d) -5	**1**

8	What will be the output of the following Python code?
	count = 10
	def update():
	   global count
	   count += 1
	update()
	print(count, end='@')
	count = 5
		1

	update() print(count) a) 11@12 b) 11@6 c) 5@6 d) 12@6	
9	Which combination of SQL commands will both change the degree and cardinality of a relation? a) ALTER TABLE and UPDATE b) UPDATE and DELETE c) ALTER TABLE and INSERT d) DROP and CREATE	**1**
10	What will be the output of the following code? L = [["Data", "Science"], ["Machine", "Learning"]] print(L[0][1][2] + L[1][0][-1]) a) ne b) In c) en d) ie	**1**

11	What will be the output of the following code?
	st = 'Computer@Science#2026'
	result = ""
	for ch in st:
	    if ch.isalnum():
	        result += ch
	print(result[::3])
	
	a) Cpece26 b) Cpecn26 c) Cmtnc26 d) Cpten06
		1

12	A relation R1 has degree 6 and cardinality 10. Another relation R2 has degree 4 and cardinality 8. If a NATURAL JOIN is performed on these relations with 2 common attributes resulting in 12 rows, what will be the degree of the resultant relation? a) 8 b) 10 c) 6 d) 4	**1**
13	Which protocol is used for secure file transfer over a network using encryption? a) FTP b) SFTP c) HTTP d) SMTP	**1**
14	Consider the given expression: print((True or False) and (False or not False) and not (True and False)) Which of the following will be the correct output? a) True b) False c) 1 d) 0	**1**
15	What will be the output of the following Python code? text = "Python Programming" print(text[-1::-2]) a) gimroPnhy b) gmrioPhny c) gmroPinyh d) gimrPhnoy	**1**

16	Which of the following is correct about the function of a Router in a computer network? a) It broadcasts data to all devices within a single network segment. b) It operates primarily at the Data Link layer to forward data frames. c) It determines the optimal path for data packets between different networks. d) It regenerates a weak signal to extend the network segment's distance.	**1**
17	Select the highest bandwidth transmission media from the followings:- a) Coaxial Cable b) Fiber Optic Cable c) Microwaves d) Radio waves	**1**
18	Complete the following SQL statement to display the vehicles registered in any district of Uttar Pradesh, using any of the option given below:- SELECT makeyear, veh_name FROM Vehicle where V_Number __________ ; a) = “UP” b) Like “UP” c) Like “UP%” d) Like “UP_ _”	**1**
19	Select the correct SQL statement from the followings which is used to delete all the records of a relation “Students” along with its structure. a) Delete ALL from Students; b) Delete * from Students; c) Drop table Students d) Drop All from Students	**1**
Q20 and Q21 are Assertion(A) and Reason(R) based questions. Mark the correct choice as: **a) Both A and R are True and R is the correct explanation for A.** **b) Both A and R are True and R is not the correct explanation for A.** **c) A is True but R is False.** **d) A is False but R is True.**		
20	**Assertion (A) :** list1.extend([4,5]) increases the length of list1. **Reason (R):**extend() adds elements one-by-one, unlike append() which adds the entire list as a single element.	**1**
21	**Assertion (A):**UNIQUE constraint allows NULL values, while PRIMARY KEY does not. **Reason (R):** A table can have multiple UNIQUE constraints but only one PRIMARY KEY.	**1**
SECTION-B (7 x 2 = 14)		
22	**A)** Differentiate between keyword and identifier with two suitable examples of each. **OR** **B)** Write short note on – a) Logical operator b) Relational operator	**2**

23 The code below is intended to remove all vowels from a string and return the count of removed vowels along with the new string. However, there are syntax and logical errors. Rewrite the code after removing all errors. Underline all corrections made. **2**

```
def remove_vowels(text)
    vowels = "aeiouAEIOU"
    new_text = ' '
    count == 0
    for char in text:
        if char not in vowel:
            new_text += char
        else:
            count += 1
    return new_text count

original = "Education"
result, num = remove_vowels(original)
print(result, num)
```

24 **A) I**. Write a statement in python to remove all whitespace characters from both ends of string s and convert it to title case. **1+1**

II. Write a one-line statement in python to extract the last 3 characters from string str and convert uppercase form.

OR

B) Using Python built-in functions only :

I) Convert list L=[1,2,3,4] → [4,3,2,1]

II) Extract substring "INFORMATICS" → "MATIC" using slicing only

25 Predict the output of the following code : **2**

```
def FUN(s):
    k=len(s)
    m=''
    for i in range(0,k):
        if s[i].isupper():
            m=m+s[i].lower()
        elif s[i].islower():
            m=m+s[i].upper()
        elif s[i].isdigit():
            m=m+s[i-1]
        else:
            m=m+'@'
    print(m)

FUN('CBSE-BOARD#2025-26')
```

26	Predict the output of the following code :
	d={'a':10,'b':20,'c':30}
	s=0
	for i in d.values():
	    if i%20!=0:
	        s+=i//10
	print(s)
		2

27	**A.** Write suitable MySQL commands for the following: i. Change the datatype of a column named phone_no in table SUPPLIER from VARCHAR(15) to INT. ii. Remove all records from a table named STUDENT but retain the table structure. **OR** **B.** Differentiate between WHERE clause and HAVING clause in SQL with suitable examples showing when each should be used.	**2**
28	**A.** Write the full forms of the following: (i) IMAP (ii) ISP **OR** **B.** Differentiate between Intranet and Internet	**1+1**
Section-C (3 x 3 = 9 Marks)		
29	**A.** Write a Python function that displays the number of lines in a text file named Quotes.txt that contain either the word "programming" or "coding" (case-insensitive). **OR** **B**. Write and call a Python function to read lines from a text file LOG.TXT and display only those lines that start with a digit (0-9).	**3**
30	A stack named **BookStack** stores book records. Each record is a List containing three elements – [BookTitle, Author, Price]. Example record: ['Python Basics', 'Guido', 499] Write the following user-defined Python functions to perform stack operations on BookStack: 1. **push_Book(BookStack, new_Book):** This function receives the stack BookStack and a new record new_Book as arguments and pushes the new record onto the stack. 2. **pop_Book(BookStack):** This function pops and returns the topmost book record from the stack.If the stack is empty, it should display "Underflow - No book to remove".	**3**

31	Predict the output of the following Python code:
	a)
	s1 = "Code#2025@Data"
	s2 = ""
	i = 0
	while i < len(s1):
	    if s1[i].isalpha():
	        s2 += s1[i].upper()
	    elif s1[i].isdigit():
	        d = int(s1[i])
	        s2 += str(d * 2)
	    elif s1[i] in "@#":
	        s2 += str(len(s2))
	    i += 1
	print(s2)
	
	OR
	b)
	data = "AI2026@ML2025@DL2024"
	parts = data.split('@')
	result = []
	for part in parts:
	    letters = ""
	    digits = ""
	    for ch in part:
	        if ch.isalpha():
	            letters += ch
	        elif ch.isdigit():
	            digits += ch
	    total = 0
	    for d in digits:
	        total += int(d)
	    if len(letters) == 2:
	        result.append(letters + str(total))
	print('#'.join(result))
		3

	Section-D (4 x 4 = 16 Marks)	
32	A table, named BOOK, in LIBRARY database, has the following structure: **Field** **Type** BNo int(4) BName varchar(15) Price float Qty int(11) Write the following Python function to perform the specified operation:	**4**

	Add() : To input details of a Book and store it in the table BOOK. **Display()**: The function should retrieve and display all records from the BOOK table where the Price is greater than 180. Assume the following for Python-Database connectivity: **Host: localhost, User: root, Password: Library**	
33	Sanya runs a stationery shop and maintains her daily transaction details in a CSV file named **Stock.csv**. The file stores details in the format: Item_ID, Item_Name, Quantity, Cost_Per_Item Write user-defined Python functions to help her manage the stock: **I. AddItem()** – To accept an item record from the user and append it to Stock.csv. Validate that Quantity > 0 and Cost_Per_Item > 0 before storing. **II. TotalCost()** – To read from Stock.csv, calculate and return the total cost of all items in stock by multiplying Quantity × Cost_Per_Item for each row.	**4**
34	Consider the table **EMPLOYEES** as given below:	**4**

Emp_ID	Name	Department	Designation	Salary	Join_Date	City
E101	Rajesh Kumar	Sales	Manager	85000	2020-03-15	Mumbai
E102	Anita Sharma	IT	Developer	75000	2021-06-20	Delhi
E103	Vikram Singh	Sales	Executive	45000	2022-01-10	Mumbai
E104	Priya Patel	HR	Manager	80000	2019-08-25	Bangalore
E105	Amit Verma	IT	Developer	72000	2021-11-30	Delhi
E106	Sneha Reddy	IT	Lead	95000	2018-05-12	Bangalore
E107	Karan Mehta	Sales	Manager	88000	2020-09-18	Mumbai
E108	Divya Nair	HR	Executive	48000	2023-02-14	Bangalore

A**. Write SQL queries for the following:**

I. Display the Name, Department and Salary of employees whose Salary is between 60000 and 90000 and working in the HR or Finance department.

II. Display each Department and number of employees in it.
Show only those departments having more than 2 employees.

III. Display Name, City and Salary increased by 8% (display only, don't update) for employees whose City starts with 'M'.

IV. List Emp_ID, Name and Salary of employees who joined after 2020 and sort salary by descending order.

OR

B. Predict the output of the following SQL queries (Assume suitable data in EMPLOYEES table):
I.SELECT Department, COUNT(*) AS EmpCount FROM EMPLOYEES GROUP BY Department HAVING COUNT(*) > 2;

II.SELECT Name, Department, Salary * 0.10 AS Bonus FROM EMPLOYEES WHERE Department = 'Sales';

III. SELECT City, MIN(Salary) AS MinSal, MAX(Salary) AS MaxSal FROM EMPLOYEES GROUP BY City;

IV. SELECT Name, Designation, Salary FROM EMPLOYEES WHERE Salary >= 80000 ORDER BY Salary DESC;

35 Meera is managing a College Database with the following tables: **4**

Table: COURSES

Course_ID	Course_Name	Department	Credits	Instructor
C101	Data Structures	CS	4	Dr. Smith
C102	Digital Logic	CS	3	Dr. Jones
C103	Thermodynamics	ME	4	Dr. Kumar
C104	Database Systems	CS	4	Dr. Smith
C105	CAD	ME	3	Dr. Sharma

Table: ENROLLMENTS

Enrollment_ID	Student_Name	Course_ID	Semester	Grade
E001	Amit	C101	III	A
E002	Priya	C101	III	B
E003	Raj	C102	II	A
E004	Amit	C104	I	A
E005	Sara	C103	II	C
E006	Priya	C104	III	A
E007	Raj	C101	IV	C

Write SQL queries for the following:
I. Write a query to change the **Credits** to 5 for the course whose **Course_ID is 'C105'**.

II. Display Student_Name, Course_Name, Grade and Semester of all students **enrolled in Semester 'III'**, using a **JOIN** between COURSES and ENROLLMENTS tables.

III. Write a query to delete all records from ENROLLMENTS where Grade is 'C'.

	IV. A. Write a query to display Student_Name, Course_Name, and Department for all students who are enrolled in any course where the Instructor name ends with 'Smith'. **OR** **B.** Display Department, total number of courses in that department, and the average credits per course. Show only those departments where the average credits are greater than 3.0.	
	Section - E (2x5 = 10 marks)	
36	Vaibhavi is studying in class 12 and trying to develop a solution for easy record handling of students' result and performance. She wants to store the following information of the students in the form of a list - Admission Number - Integer Name - String Percentage - float Her teacher guided her to use binary files for this purpose and create the following user defined function - **(a)** A Function **ADD_Data()** to input the records and add it to a Binary file **Student.Dat.** **(b) CountRec()** - Accepts the range of the percentage, display the records and count of the records lying in the given range.	**5**
37	National Centre for Indigenous Arts has its Headquarter in Dehi and just set up a new Campus in Kolkata. They want to set up a Local area network in Kolkata campus. The Campus has four buildings : ● Main Building ● Admin ● Finance ● Academic The distances between various buildings of university are given as: **Block to Block Distance in Meters**	**5**

FROM	**TO**	**Distance (in Meters)**
Main	Admin	50
Main	Finance	100
Main	Academic	70
Admin	Finance	50
Finance	Academic	70
Admin	Academic	60

Number of Computers in Each Building

Building Name	**Number of Computers**
Main Building	80
Admin Building	75
Finance Building	50
Academic Building	60

	As a network expert, you are required to give best possible solutions for the given queries of the university administration:- **(a)** Suggest the most appropriate location of the server inside the campus. Justify your choice. **(b)** Which hardware device will you suggest to connect all the computers within each building? **(c)** Draw the cable layout to efficiently connect various buildings within the campus. Which cable would you suggest for the most efficient data transfer over the network? **(d)** Is there any requirement of a repeater in the given cable layout? Justify your answer. **(e)** What would be your recommendation for enabling live visual communication between the Admin Office at the Kolkata campus and the DELHI Headquarter from the following options: a) Video Conferencing b) Email c) Telephony d) Instant Messaging **OR** What type of network (PAN, LAN, MAN, or WAN) will be formed between Delhi Headquarter and Kolkata Campus?	